

Multilevel Models

Last Time

Our future AI overlords, and how they can be help us in the present

Today

Multilevel Models

- Why are they?
- What are they?
- How do I?

Why MLM?

- MLM is used when data are non-independent.
- Non-independence means that knowing one observation on the outcome tells you something about another observation.
- Happens when data have a clustered or nested structure.

Why MLM?

Examples of *clustered* or *nested* data

- Students nested in classrooms
- Parents and children nested in families
- Partners nested within couples
- Neighborhoods nested within cities
- Time points nested within people
- In experimental settings: stimuli nested within people, but also people nested within stimuli

Violating the independence assumption is bad.

$$Y_i = b_0 + b_1 X_1 + \epsilon_i$$

- Values of Y should be uncorrelated with each other. If not, then there will be correlations (dependencies) among the residuals.
- Leads to you to *underestimate* standard errors.
- **INFLATED TYPE I ERROR**

One option is to aggregate (discard within-cluster variability)

- e.g., if you have 3 timepoints for each person, average across those and use peoples' average levels as the outcome.

Another option is to cancel out the cluster variable and only look at the within-level (discard between-cluster variability)

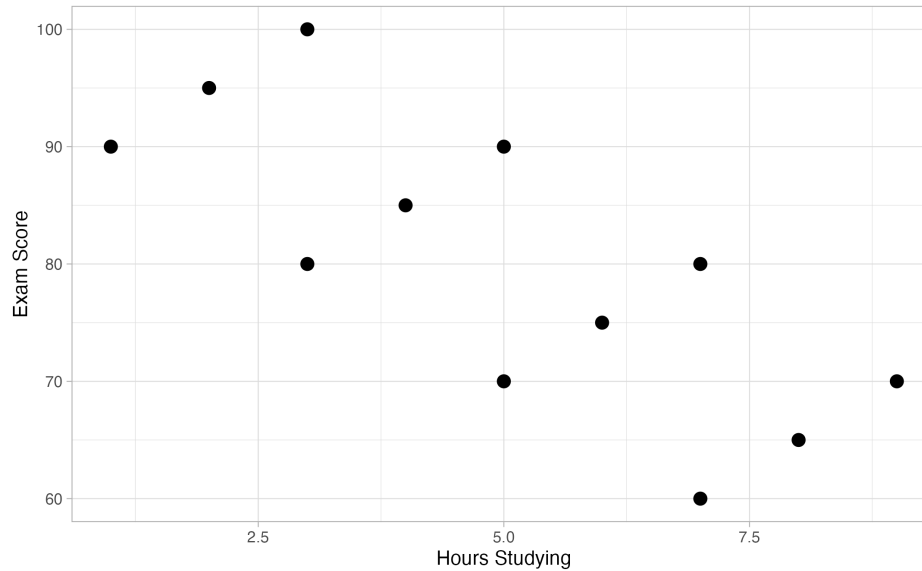
- e.g., remove variability due to average levels, and use peoples' deviations across the three time points as the outcome.
- This is basically repeated measures ANOVA

Both options are terrible.

Obscures different meanings at different levels of analysis.

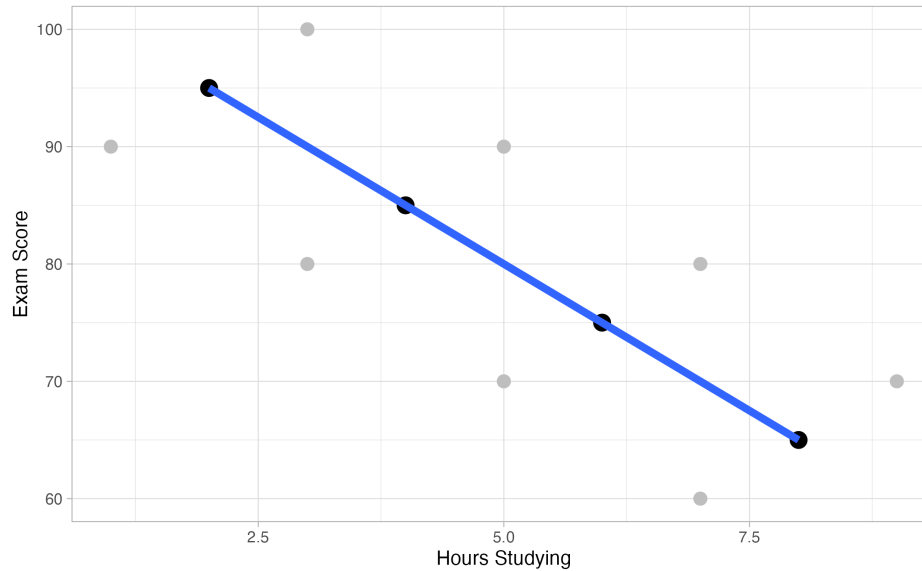
Obscures different meanings at different levels of analysis.

- e.g., does studying $\rightarrow$ better grades?



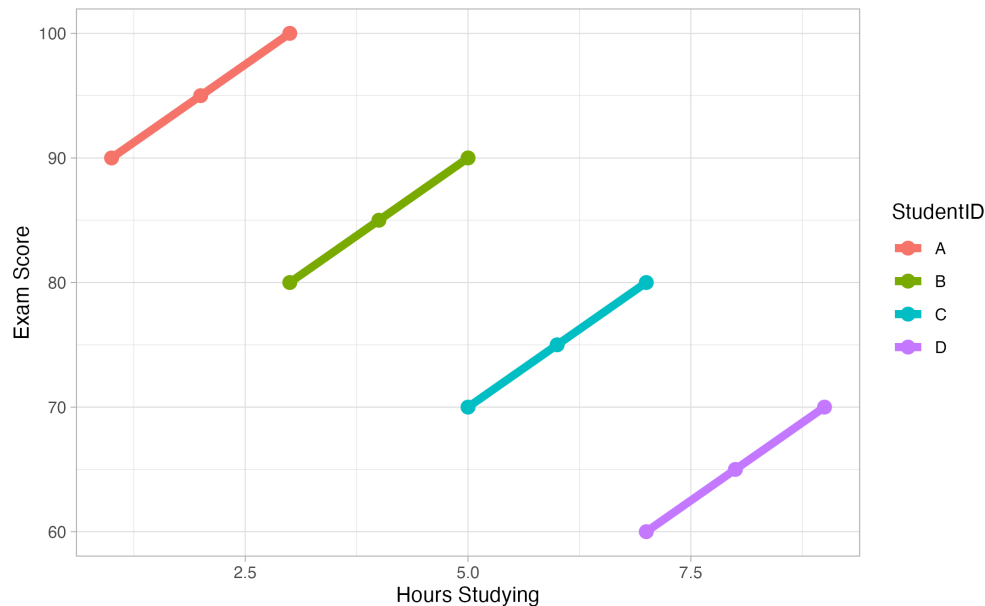
Obscures different meanings at different levels of analysis.

- e.g., does studying $\rightarrow$ better grades?



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- e.g., does studying $\rightarrow$ better grades?



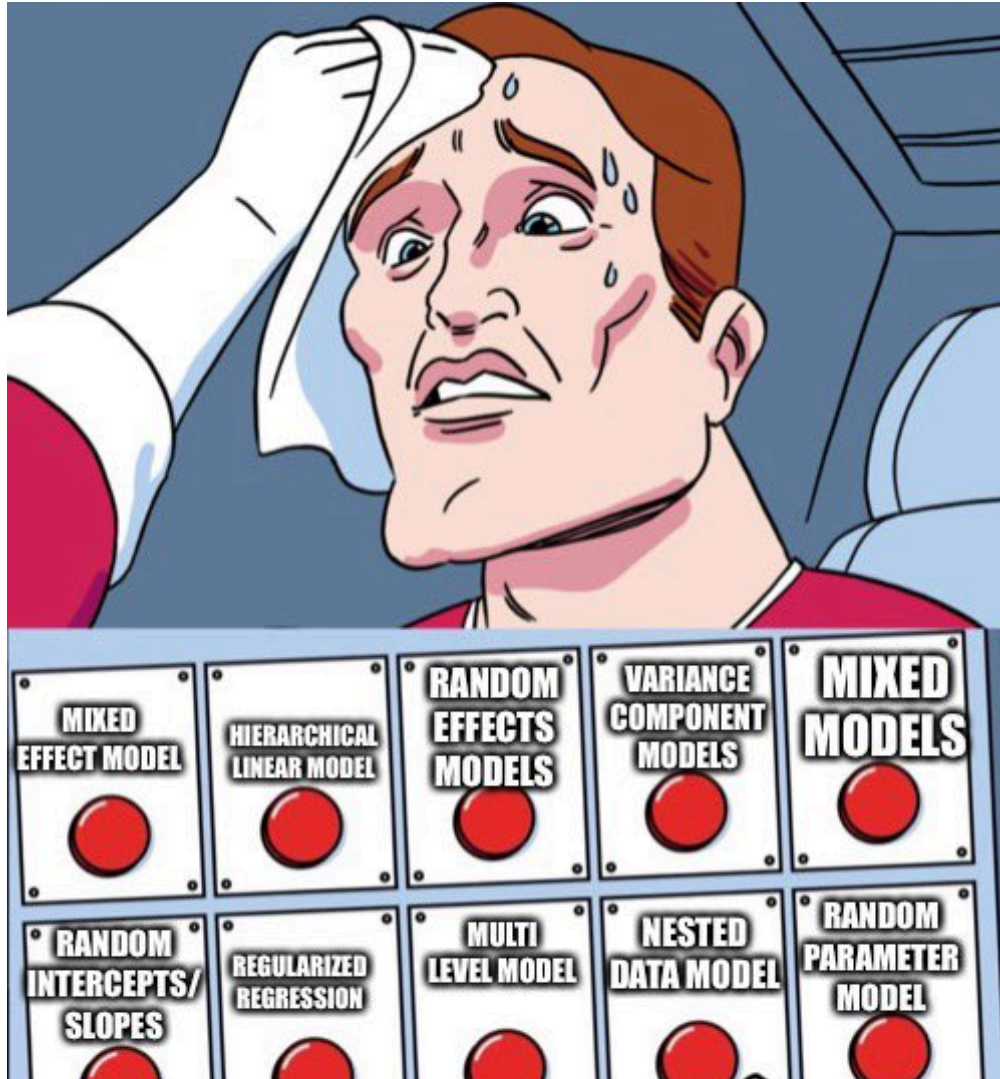
Both options are terrible.

Obscures different meanings at different levels of analysis.

- Between person: do people who study more get better grades (vs. people who study less)?
- Within person: when a person studies more, do they get better grades (vs. when they study less)? Neither option really allows for people to have individual differences.
- i.e., assume that the effect is the same for everyone.

Multilevel Models: **the good option**

What is MLM?



What is MLM?

- MLM is going to directly model the dependencies by decomposing variance into within- vs. between-cluster variance.
- Separate effects into Level 1 (within-cluster) and Level 2 (between-cluster)

The Null model

What is the null model in regular linear regression?

$$Y_i = b_0 + \epsilon_i$$

$$\hat{Y}_i = \bar{Y}_i$$

- Now we are in multilevel land. We are going to separate this into two equations.
- decompose the variance into Level 1 (within) and Level 2 (between)

The Null Model

Level 1

$$Y_{ij} = b_{0j} + \epsilon_{ij}$$

- i = observations
- j = clusters
- b_{0j} = mean level of Y for each cluster j
- ϵ_{ij} = within cluster variance around each b_{0j}

The Null Model

Level 1

$$Y_{ij} = b_{0j} + \epsilon_{ij}$$

Level 2

$$b_{0j} = \gamma_{00} + U_{0j}$$

- j = clusters
- γ_{00} = the grand mean of Y across all clusters
- U_{0j} = variance of the cluster means b_{0j} around the grand mean γ_{00}

The Null Model

Level 1

$$Y_{ij} = b_{0j} + \epsilon_{ij}$$

Level 2

$$b_{0j} = \gamma_{00} + U_{0j}$$

Can simplify into a single formula

$$Y_{ij} = \gamma_{00} + U_{0j} + \epsilon_{ij}$$

The Null Model

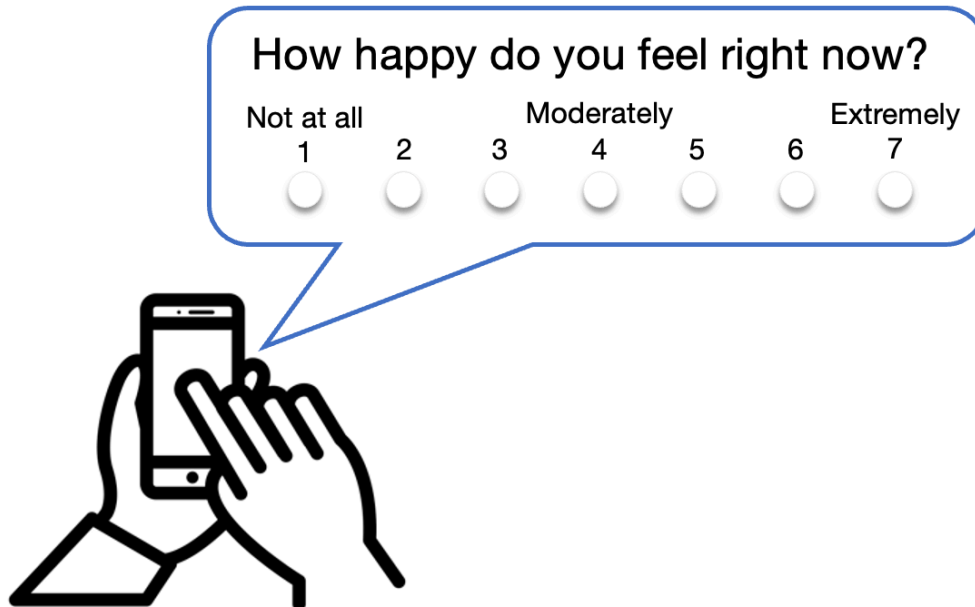
$$Y_{ij} = \gamma_{00} + U_{0j} + \epsilon_{ij}$$

Fixed vs. Random effects

- U_0 is a random effect, meaning that it varies across clusters (j)
- γ_{00} is a fixed effect, meaning it is the same across all clusters.

Research example

EMA study, we have 6 people who reported how happy they feel 6x per day for 10 days



Research example

EMA study, we have 6 people who reported how happy they feel 6x per day for 10 days

```
dat <- read.csv("pa_esm.csv")
```

```
##      ID posemo  obs
## 1  Shelly    10    1
## 2    Ben     5    1
## 3  Ethan     7    1
## 4   Josh     7    1
## 5   Qilin     8    1
## 6  Jessie     8    1
## 7  Shelly    10    2
## 8    Ben     3    2
## 9  Ethan     7    2
## 10   Josh     4    2
## 11  Qilin     5    2
## 12  Jessie     7    2
##
## # 1-12 of 360 rows
```

Research example

EMA study, we have 6 people who reported how happy they feel 6x per day for 10 days

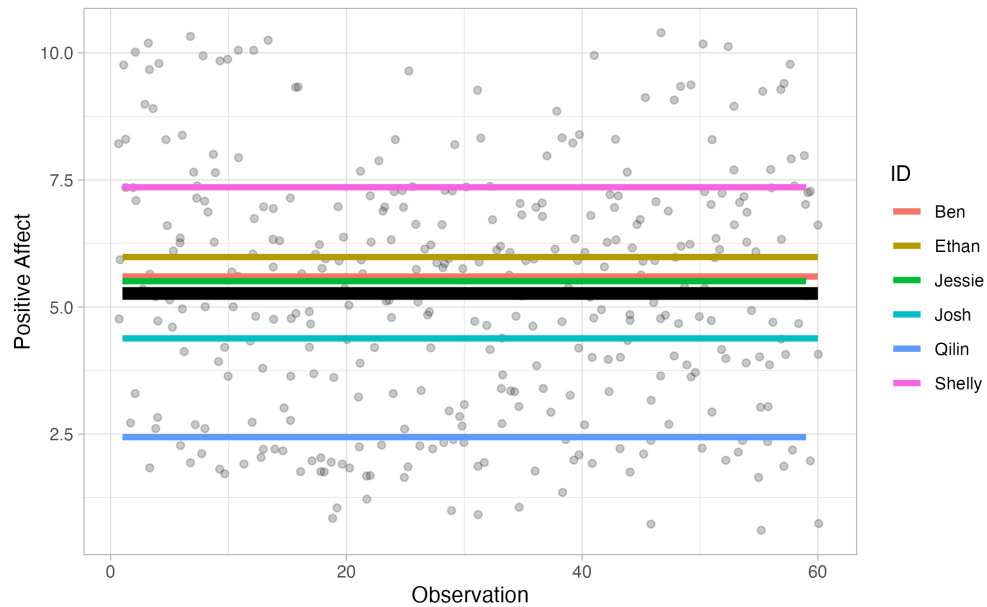
Null Model

$$Y_{ij} = \gamma_{00} + U_{0j} + \epsilon_{ij}$$

$$\begin{aligned} \text{Happy}_{ij} = & \text{Grand Mean(Happy)} + \\ & \text{Person Mean(Happy)}_j - \text{Grand Mean(Happy)} + \\ & \text{Happy}_{ij} - \text{Person Mean(Happy)}_j \end{aligned}$$

Research example

Null Model



- this should look familiar?

Intraclass correlation

$$\text{ICC} = \frac{U_{0j}}{U_{0j} + \epsilon_{ij}}$$

- Out of the total variation, how much is due to between-cluster differences?
- Metric of dependency: how well can you predict a persons score at one moment from their score at another moment?
- Also has a substantive meaning: what % of var in happiness is explained by individual differences in average happiness vs. people varying in happiness from moment to moment?

Adding a level-1 predictor

- Correlating a repeated outcome with something else that is measured repeatedly within the cluster.
- For example, do people feel happier during moments when they are acting more extraverted?

Random Intercept model

Level 1

$$Y_{ij} = b_{0j} + b_1 X_{ij} + \epsilon_{ij}$$

Level 2

$$b_{0j} = \gamma_{00} + U_{0j}$$

$$b_1 = \gamma_{10}$$

Simplifies to

$$Y_{ij} = \gamma_{00} + \gamma_{10} X_{ij} + U_{0j} + \epsilon_{ij}$$

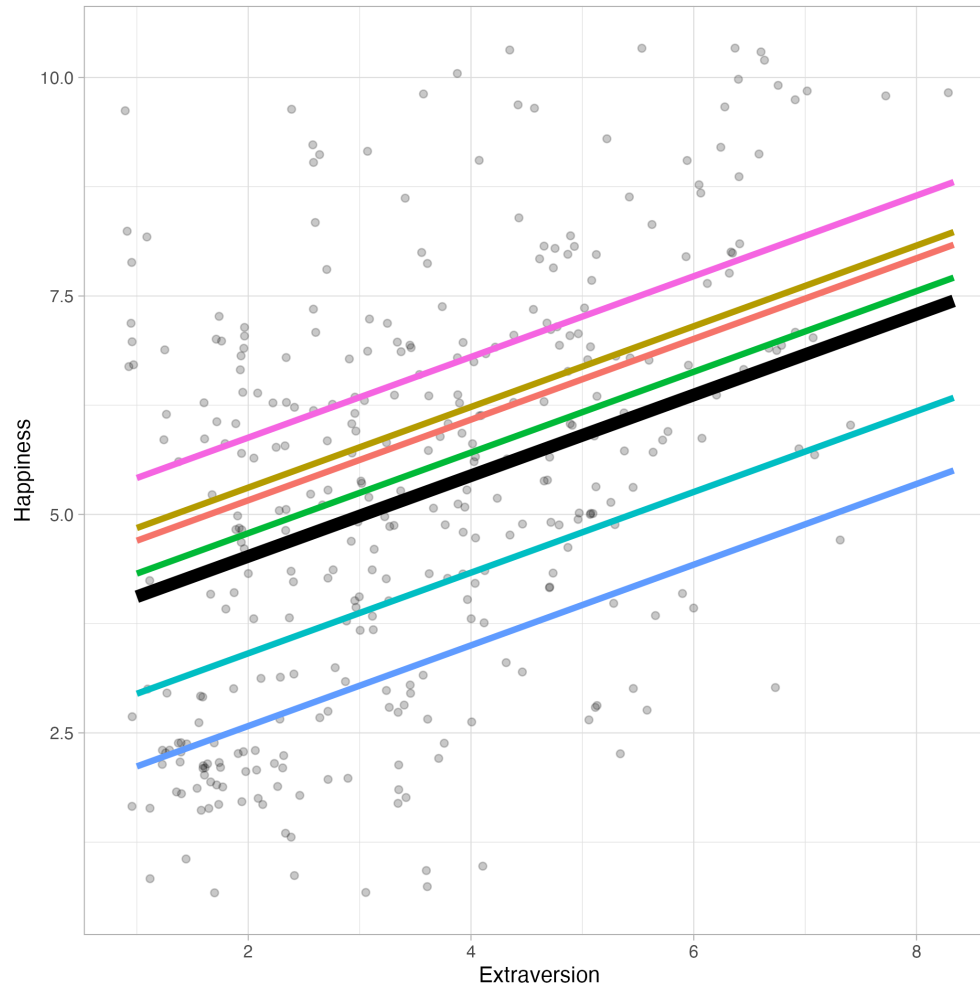
Random Intercept model

$$Y_{ij} = \gamma_{00} + \gamma_{10}X_{ij} + U_{0j} + \epsilon_{ij}$$

Two fixed effects

- γ_{00} = fixed intercept (average persons level of Y when $X = 0$)
- γ_{10} = fixed slope (expected change in Y for 1-unit change in X) One random effect
- U_{0j} = individual differences in the intercept
- No random effect for slope.
- Thus, the slope is *fixed*, meaning our model says everyone has the same effect.

Random Intercept model



Random Slope Model

- The random intercept model incorporates individual differences by saying that each person gets their own intercept.
- Incorporates individual differences in average levels or starting levels.
- But we can go further.
- People might not only have different means, they may also have different relationships between X and Y .

Random Intercept model

Level 1

$$Y_{ij} = b_{0j} + b_1 X_{ij} + \epsilon_{ij}$$

Level 2

$$b_{0j} = \gamma_{00} + U_{0j}$$

$$b_1 = \gamma_{10}$$

Simplifies to

$$Y_{ij} = \gamma_{00} + \gamma_{10} X_{ij} + U_{0j} + \epsilon_{ij}$$

Random Slope Model

Level 1

$$Y_{ij} = b_{0j} + b_1 X_{ij} + \epsilon_{ij}$$

Level 2

$$b_{0j} = \gamma_{00} + U_{0j}$$

$$b_1 = \gamma_{10} + U_{1j}$$

Simplifies to

$$Y_{ij} = \gamma_{00} + \gamma_{10} X_{ij} + U_{0j} + U_{1j} X_{ij} + \epsilon_{ij}$$

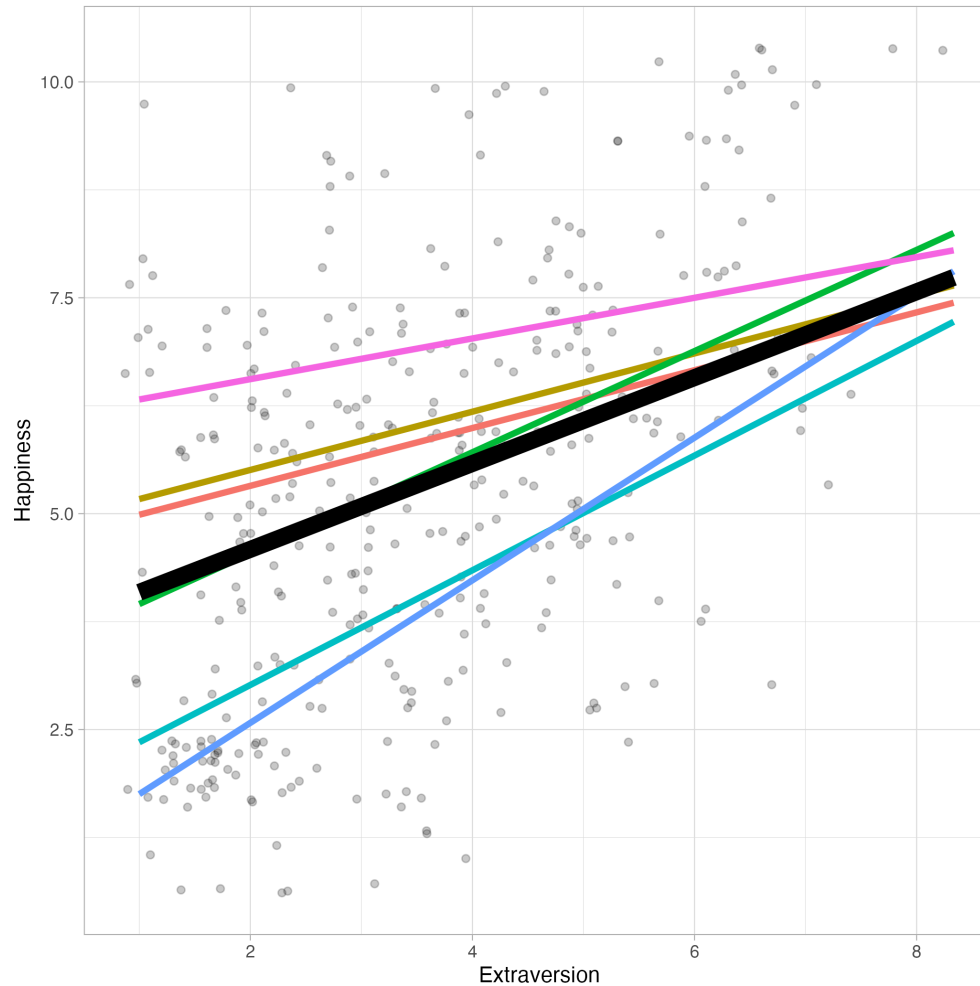
Random Slope model

$$Y_{ij} = \gamma_{00} + \gamma_{10}X_{ij} + U_{0j} + U_{1j}X_{ij} + \epsilon_{ij}$$

Two fixed effects

- γ_{00} = fixed intercept (average person's level of Y when $X = 0$)
 - γ_{10} = fixed slope (average person's change in Y for 1-unit change in X)
- Two random effects
- U_{0j} = individual differences in the intercept
 - U_{1j} = individual differences in the slope
- Every person gets their own intercept (b_{0j})
+ slope (b_{1j})

Random Slope model



Technical things to be aware of

Estimation

- Do you remember maximum likelihood estimation?
- e.g., from logistic?
- Multilevel models use MLE, not OLS

Partial pooling

- Each cluster/group/person is treated as coming from a population.
- All clusters are alike in some ways, but also different in their own unique ways.
- With MLM, we fit a regression separately for each cluster, but we are not just getting the best fit within the cluster.
- We adjust predictions using information from other groups. This helps us avoid overfitting.
- Total pooling = everyone is the same.
- No pooling = calculate each person's fit in isolation.

Practical considerations

Centering

- Centering is really important in MLM.
- As a rule, any given level-1 predictor is really 2 predictors (one per each level)
- You want to separate the within-sources of variance from the between-sources. If you don't level 1 and level 2 get "smushed" together.

Group-mean Centering

- To isolate level 1 effects, we typically want to create group-mean centered variables (AKA person-mean centered if people are level 2)
- Each cluster at level 2 gets its own mean, then we subtract that mean from each observation for that cluster.
- Eliminates between-cluster differences in the predictor variable.

Group-mean Centering

```
# Person-mean center extraversion
dat <- dat %>%
  group_by(ID) %>%
  mutate(extraversion.m = mean(extraversion, na.rm = T),
         extraversion.c = extraversion - extraversion.m)
```

```
## # A tibble: 360 × 4
## # Groups:   ID [6]
##   ID      extraversion extraversion.m extraversion.c
##   <chr>          <dbl>          <dbl>          <dbl>
## 1 Ben            2.33            2.91          -0.572
## 2 Ben            2.67            2.91          -0.239
## 3 Ethan          4            3.41           0.589
## 4 Ethan          4            3.41           0.589
## 5 Josh           3.67            4.18          -0.517
## 6 Josh           2.33            4.18          -1.85
## 7 Qilin          5            1.86           3.14
## 8 Qilin          1.67            1.86          -0.190
## 9 Shelly         6.67            5.10           1.57
## 10 Shelly        7            5.10           1.90
## # i 350 more rows
```

Power

- There are many levels of effects, so power depends on what level you care about.
- e.g., level 1 effects depend on number of observations per cluster. Level 2 effects depend on the number of clusters.
- Because MLM effects include many different components of variance and covariance, power analyses are much less straightforward.
- Need to run simulations.
- PowerAnalysisIL shinyapp (Lafit et al., 2021, AMPPS):

Preregistration

Modeling choices can influence the results! So you should be explicit about them.

How are predictors centered? What random slopes will you model? What will you do if the model doesn't converge?

Convergence issues

- Models can easily become complex -> estimation issues (e.g., too many random slopes, too little variance)
- Which random slopes *should* you fit?

What can you do in cases of non-convergence?

- Adjust the optimizer
- Drop random slopes with little variance
- Bayes estimators can fit more complex models

Running a *MLM* in R

Null Model

```
library(lme4)
mod.null <- lmer(posemo ~ (1 | ID),
                 data = dat)
summary(mod.null)
```

```
## Random effects:
```

```
##   Groups      Name          Variance Std.Dev.
##   ID          (Intercept)  2.690     1.640
##   Residual                        3.474     1.864
##
```

```
## Fixed effects:
```

```
##              Estimate Std. Error    df t value
## (Intercept)   5.2134    0.6771 4.9817  7.699
```

Null Model

Calculating the intraclass correlation

```
null_var <- data.frame(VarCorr(mod.null), row.names = c('betw  
ICC <- null_var['between', 'vcov']/(null_var['between', 'vcov  
ICC
```

```
## [1] 0.4365
```

$$ICC = \frac{U_{0ID}}{U_{0ID} + \epsilon_{IDmoment}}$$

- 43% of variation in happiness is due to individual differences in average happiness
- 57% of variation is due to people changing in happiness moment-to-moment

Random Intercept Model

```
mod.fixed <- lmer(posemo ~ extraversion.c + (1 | ID),  
                  data = dat)  
summary(mod.fixed)
```

```
## Random effects:
```

##	Groups	Name	Variance	Std.Dev.
##	ID	(Intercept)	2.699	1.643
##	Residual		3.126	1.768

```
## Number of obs: 345, groups: ID, 6
```

```
##
```

```
## Fixed effects:
```

##		Estimate	Std. Error	df	t value
##	(Intercept)	5.21301	0.67749	4.98353	7.695
##	extraversion.c	0.44867	0.07209	337.98351	6.224

Random Slope Model

```
mod.random <- lmer(posemo ~ extraversion.c + (extraversion.c
                  data = dat)
summary(mod.random)
```

```
## Random effects:
```

```
##   Groups      Name                Variance Std.Dev.  Corr
##   ID          (Intercept)         2.7052   1.6448
##               extraversion.c 0.1066    0.3264   -0.94
## Residual                        3.0412    1.7439
## Number of obs: 345, groups:  ID, 6
```

```
##
```

```
## Fixed effects:
```

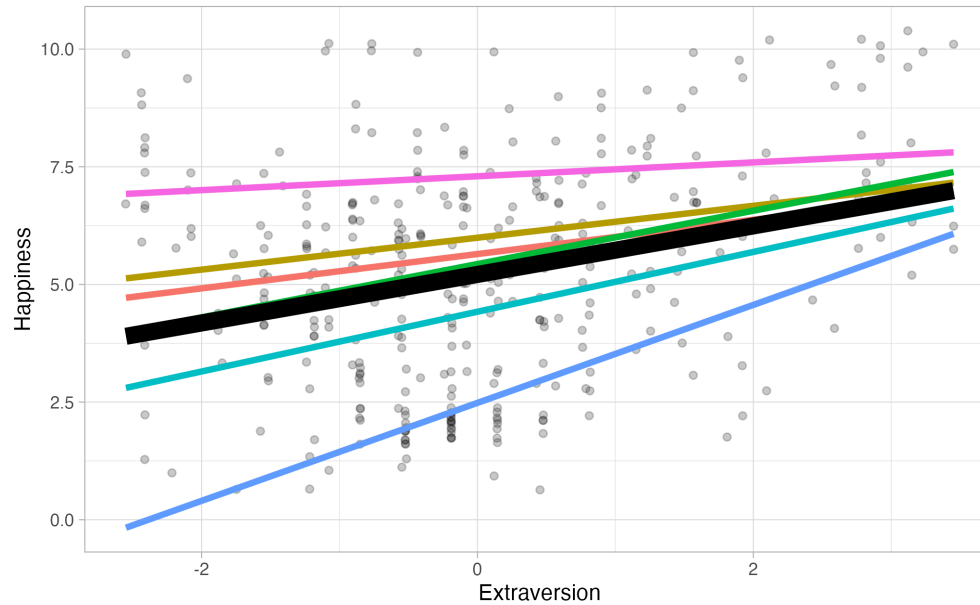
```
##               Estimate Std. Error    df t value
## (Intercept)      5.2118     0.6780 4.9838   7.687
## extraversion.c    0.5156     0.1525 3.5774   3.380
```

Random effects: variances and covariances

```
## # A tibble: 6 × 3
##   id          U0      U1
##   <chr>      <dbl> <dbl>
## 1 Shelly    0.460 -1.10
## 2 Ben      0.828 -0.484
## 3 Ethan    0.234 -0.443
## 4 Josh    -0.840  0.253
## 5 Qilin   -2.89   0.328
## 6 Jessie   2.21   1.44
```

(r) = -.94 means that people who have higher average levels of happiness have a less strong association between momentary extraversion and momentary happiness.

Random Slope Model



Random Slope Model

Can think of each person's predicted scores as comprising the overall, fixed component + their own unique random component

$$\hat{Y}_{iShelly} = \gamma_{00} + \gamma_{10}X_{ij} + U_{0Shelly} + U_{10Shelly}X_{ij}$$

$$\begin{aligned} \hat{Happy}_{iShelly} = & 5.25 + .51 * Extra_{ij} + .46 + \\ & -1.10 * Extra_{ij} \end{aligned}$$

Infinite possibilities

Adding Level-2 predictors

- These can include cross-level interactions where you predict individual differences in slope from individual differences in measured variables at level 2.
- e.g., do more extraverted people get a bigger happiness boost from being extraverted in the moment?

Multiple levels of nesting (level 3, level 4...)

- e.g., students within classrooms, within schools, within districts...

Infinite possibilities

Clusters can be crossed rather than nested

- e.g., in experiments
- The same person responds to many stimuli.
- The same stimuli receives responses from many people.
- But one is not nested in the other.

Wrap-up

- You should be using multilevel models!
- Applicable anytime that the same person does multiple things or a few people who have something in common do anything.
- You should take HLM with Emily Willroth in the Fall!

Next Time

Bayes: Josh Jackson Returns

Final class of the semester next Tuesday :0